Contents

Edexcel IAL Pure Mathematics 1 (WMA11/01) — Summer 2022 Paper Review

Recall & Jargon Pack

Tutorial: Student × Tutor · recorded 16 June 2026 · paper reviewed = **WMA11/01, Summer 2022** (paper code P69458A, 75 marks, 10 questions).

This pack is a fast-recall companion to the full revision guide (`guide_v2/revision_guide.pdf`, which passed independent AV audit v1). It compresses every term, formula and method touched in the lesson into paste-ready tables, built to keep the **evidence boundaries visible at every line** — you should never have to guess whether a claim comes from the official paper or from the video.

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0. How to read this pack — provenance & status boundaries

0.1 Where each fact comes from (authority order)

Source	Role in this pack	Authority
`guide_v2/revision_guide.pdf`, `revision_guide_text.md`	Question wording, marks, answers	**Authoritative** — evidence-led rebuild, AV-audit PASS
`guide_v2/source_register.md`	Official Pearson QP/MS identity, hashes	**Authoritative** — source provenance
audio transcript `loom_transcript_timestamped.json`	Student's spoken admissions, chronology	**Primary (lesson)** — one-sided audio
AV reconciliation + independent AV audit v1	Per-question review status, score reconciliation	**Primary (evidence)**
Gemini AV ledger	Board scene reads	**Visual lead only** — never source authority

Rule applied throughout (QP/MS vs video): the official WMA11/01 Summer-2022 QP/MS control all question wording, marks and final answers. The video/audio are used only for *method, diagnosis of the Student's work, and exam-technique advice*. The legacy "29/75" cover total is **not** reconciled — see §5.

> ⚠ **No verbatim Tutor quotes.** The audio is one-sided (Student's microphone); the Tutor's speech is largely inaudible. The one retained Q1 board annotation is a *written* note, labelled as such, not a spoken quote.

0.2 Status legend (used in the "Lesson status" column of §9)

Tag	Meaning
Reviewed & corrected	Worked live in the lesson; mistake diagnosed and fixed
Reviewed	Worked through live; method confirmed
Touched briefly	Mentioned / re-explained, not a logged lost-mark review
NOT reviewed live	Printed on the shared worksheet but never solved; scored state unknown — **reference only**

0.3 Evidence-tier legend

`[OFFICIAL]` Pearson QP/MS (highest) · `[AUDIO]` confirmed in timestamped audio · `[VISUAL-LEAD]` automated board read, lead only · `[GUIDE-CLAIM]` legacy guide claim, not independently confirmable · `[UNAVAILABLE]` not present in any retained artifact (preserved as a gap).

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1. Recall tables — terms & jargon by topic

Columns: **Term · Student-friendly meaning · Exact mathematical meaning · Symbols/conditions · What it is NOT · Lesson example · Common trap.**

1.1 Indefinite integration (Q1)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Power rule (integrate)	Raise the power, divide by the **new** power	$\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$	$n \neq -1$; add c	Not differentiation (drop power by one)	Q1: $\int(10x^5 + 6x^3 - 3x^{-2})dx$
Rewrite negative powers	Turn $3/x^2$ into $3x^{-2}$ before integrating	$\frac{a}{x^k} = ax^{-k}$	$k > 0$	Not leave as a fraction	Q1 third term: $-3x^{-2} \to +3x^{-1} = +3/x$
"Add one to the power"	New exponent is old $+1$	$x^{-2} \to x^{-1}$	—	Not x^{-3}	Q1: $+3/x$ term

1.2 Sine rule & obtuse angles (Q2)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Sine rule	Ratio of side to sine of opposite angle	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	Labelled triangle	Not cosine rule	Q2: $\sin x = \frac{21 \sin 25^\circ}{13} = 0.6827$
Two solutions of sine	$\sin\theta = \sin(180^\circ - \theta)$	Principal value + its supplement	For $0^\circ < x < 180^\circ$	Not only the acute angle	Q2 obtuse: $x = 180^\circ - \arcsin(0.6827) = 136.95°$
Accuracy to 4 d.p.	Keep four decimal places through the sine value	Carry full precision, round at the end	Stated "4 d.p."	Not 2 or 3 d.p.	Q2(a): 0.6827 then 136.95°

1.3 Surds (Q3)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Simplify a surd quotient	Split the fraction, simplify each root	$\frac{\sqrt{180}-\sqrt{80}}{\sqrt{5}} = \frac{\sqrt{180}}{\sqrt{5}} - \frac{\sqrt{80}}{\sqrt{5}}$	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{a/b}$	Not $\sqrt{a-b}$	Q3(i): $= 6 - 4 = 2$
Rationalise the denominator	Remove the surd from the bottom	Multiply top and bottom by the conjugate	Conjugate of $a + b\sqrt{5}$ is $a - b\sqrt{5}$	Not just multiply by the same surd	Q3(ii): $\frac{4\sqrt{5}-5}{7-3\sqrt{5}} = \frac{25}{4} + \frac{13}{4}\sqrt{5}$

1.4 Graph transformations (Q4)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Vertical shift $f(x) + k$	Move the graph up/down	$y = f(x) - 2$ shifts down 2	y-values change by k	Not a horizontal move	Q4(i): min $(-1, -2)$, max $(3/2, 0)$, asymptote $y = -1$
Reflection $f(-x)$	Mirror in the y**-axis**	$x \to -x$, y unchanged	Horizontal only	Not $f(x)$ reflected, not a vertical flip	Q4(ii): min $(1, 0)$, max $(-3/2, 2)$, asymptote $y = 1$ unchanged
"Asymptote unchanged"	A y-axis reflection leaves horizontal lines put	$y = 1$ stays $y = 1$	Horizontal asymptote	Not flipped with the curve	Q4(ii)

1.5 Completed square / lines / regions (Q5)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Completed-square (quadratic)	Turn $ax^2 + bx + c$ into vertex form	$12 - \frac{4}{3}(x+2)^2 = -\frac{4}{3}(x-1)(x+5)$	Keep the leading coefficient	Not $12 - (x+2)^2$ (drops the $4/3$)	Q5(a): vertex form + factorised form
Equation of a line	$y = mx + c$ form	Q5(b): $\ell_2 : y = -\frac{5}{4}x - \frac{25}{4}$	Use gradient + point	Not just the gradient	Q5(b)
Region from inequalities	Shade where all inequalities hold	Q5(c): three inequalities define the region	Test a point	Not one inequality alone	Q5(c)

1.6 Simultaneous equations → quartic (Q6)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Hidden quadratic	Substitute $u = x^2$ to drop the degree	$2x^4 - 15x^2 - 50 = 0 \Rightarrow 2u^2 - 15u - 50 = 0$	$u = x^2 \geq 0$	Not a genuine quartic to factor directly	Q6
Reject negative root	A square cannot be negative	Reject $u = -5/2$	$x^2 \geq 0$	Not keep both u values	Q6: $x^2 = 10$
Exact surd answers	Leave roots unsimplified	$(\pm\sqrt{10}, \pm 4\sqrt{10})$	Pair x and y	Not decimals	Q6

1.7 Differentiation: find A, then $f(x)$ (Q7)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Differentiate fractional powers	$x^{1/2} \to \frac{1}{2}x^{-1/2}$; $x^{-2} \to -2x^{-3}$	$f'(x) = 2x^{-1/2} + Ax^{-2} + 3$	Keep exact	Not decimal approximations	Q7(a): $f''(4) = 0 \Rightarrow A = -4$
Method must justify answer	A right number from invalid working scores 0	Validate each step	MS awards M marks per valid step	Not “answer only”	Q7(a)
Integrate to recover f	Reverse each term, add c, use a point	$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}$	Use $f(12) = 8\sqrt{3}$	Not forget the constant	Q7(b)

1.8 Sector / arc / perimeter / area (Q8)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
Arc length	$s = r\theta$ (radians)	Q8(a): arc CD $= 35.9$ cm	θ in radians	Not $r\theta$ in degrees	Q8(a): $\theta = 5/3$
Perimeter of sector	Two radii + arc	Q8(b): perimeter $= 603$ cm	Add both radii	Not arc only	Q8(b)
Sector area	$A = \frac{1}{2}r^2\theta$	Q8(c): area ≈ 4730 cm² $= 0.473$ m²	Convert cm²→m² by $\div 10000$	Not cm² stated as m²	Q8(c)

1.9 Trig: values, sin 2x, solve sin 2x = p (Q9)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
$\sin(180^\circ - \alpha)$	$\sin\alpha$ (symmetry)	“equal to negative sine alpha”: $\sin(180^\circ - \alpha) = \sin\alpha$	Quadrant II symmetry	Not $-\sin\alpha$	Q9(a): $-\sin\alpha = -p$
$y = \sin 2x$ period	Half the period of $\sin x$	Period $= 180^\circ$ (horizontal half-stretch)	Inside factor 2	Not 360°	Q9(b)
Solve $\sin 2x = p$	One root $x = \alpha/2$; partner by symmetry	$x = \alpha/2$ and $x = 90^\circ - \alpha/2$	Both in range	Not one solution only	Q9(c)

1.10 Normal to a curve, find k (Q10)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example
$\frac{dy}{dx}$ from powers	$x^n \to nx^{n-1}$	Q10(a): $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$	—	Not integrate	Q10(a)
Normal gradient	Negative reciprocal of tangent	$m_{\text{normal}} = -1/(dy/dx)$	Perpendicular	Not the tangent slope	Q10(b): $12x^2 + 4x - 33 = 0$
Solve for k	Use the normal condition at a point	Q10(c): $x = 3/2$; (d) $k = 5/4$	Substitute the point	Not guess	Q10(d)

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2. Cross-tab: Integration

Method	Formula	Use when…	This paper	Trap
Integrate power	$\int x^n dx = \frac{x^{n+1}}{n+1} + c$	Powers of x	Q1	Differentiate instead of integrate
Negative power	$-3x^{-2} \to +3x^{-1}$	Term with x below the line	Q1 third term	Sign on the new negative power

3. Cross-tab: Sine rule

Step	Statement	This paper	Trap
Rule	$\frac{a}{\sin A} = \frac{b}{\sin B}$	Q2	Cosine rule by mistake
Two angles	$\sin\theta = \sin(180^\circ - \theta)$	Q2 obtuse $x = 136.95^\circ$	Pick acute when obtuse needed
Precision	Carry 4 d.p.	Q2(a) 0.6827	Round early → accuracy mark

4. Cross-tab: Surds

Step	Statement	This paper	Trap
Split	$\frac{\sqrt{a}-\sqrt{b}}{\sqrt{c}} = \frac{\sqrt{a}}{\sqrt{c}} - \frac{\sqrt{b}}{\sqrt{c}}$	Q3(i)=2	Don't split denominator only
Rationalise	Multiply by conjugate	Q3(ii) $25/4 + 13\sqrt{5}/4$	Conjugate sign error

5. Cross-tab: Graph transformations

Transform	Effect	This paper	Trap
$f(x)-2$	Down 2	Q4(i) asymptote $y=-1$	—
$f(-x)$	Reflect in y-axis	Q4(ii) asymptote $y=1$ unchanged	Moving the horizontal asymptote

6. Cross-tab: Hidden quadratic

Step	Statement	This paper	Trap
Substitute $u=x^2$	Quartic → quadratic	Q6	Forget $u=x^2$
Reject $u<0$	$x^2\geq 0$	Q6 reject $-5/2$	Keep negative root
Pair surds	Exact (x,y)	Q6 $(\pm\sqrt{10},\pm 4\sqrt{10})$	Decimal/round

7. Cross-tab: Differentiation

Step	Statement	This paper	Trap
Fractional powers	$x^{1/2}\to\frac{1}{2}x^{-1/2}$	Q7(a) $A=-4$	Dropped power
Method scores	Valid steps only	Q7(a)	Right answer, wrong route = 0
Integrate back	Use a point for c	Q7(b) $-109/3$	Rounded constant

8. Cross-tab: Trigonometry (Q9)

Fact	Statement	This paper	Trap
Symmetry	$\sin(180^\circ-\alpha)=\sin\alpha$	Q9(a) $-p$	Sign
Period	$y=\sin 2x$ period 180°	Q9(b)	Leave 360°
Roots	$x=\alpha/2,\ 90^\circ-\alpha/2$	Q9(c)	One root only

9. Per-question quick-recall map (status + one-line takeaway)

Wording from the **official QP**; answers from the **MS**; lesson takeaway from the **AV-reconciled guide**. `[NOT reviewed live]` = printed on the shared worksheet, never solved; scored state unknown.

Q (marks)	Official task (QP)	Answer (MS)	Status	One-line takeaway
1 — 4	$\int(10x^5+6x^3-3/x^2)dx$	$\frac{5}{3}x^6+\frac{3}{2}x^4+\frac{3}{x}+c$	Reviewed & corrected	Integrate, don't differentiate; watch the $-3x^{-2}$ sign

Q (marks)	Official task (QP)	Answer (MS)	Status	One-line takeaway
2 — 5	Triangle: $AB = 21, BC = 13, \angle BAC = 25°$; find sin x, then obtuse x	sin $x = 0.6827$; $x = 136.95°$	Reviewed	Carry 4 d.p.; pick the obtuse supplement
3 — 5	(i) $\frac{\sqrt{180}-\sqrt{80}}{\sqrt{5}}$; (ii) rationalise $\frac{4\sqrt{5}-5}{7-3\sqrt{5}}$	(i) 2; (ii) $\frac{25}{4} + \frac{13}{4}\sqrt{5}$	Touched briefly	Split the numerator; conjugate with correct sign
4 — 6	Sketch $f(x) - 2$ and $f(-x)$ (min $(-1, 0)$, max $(3/2, 2)$, asymptote $y = 1$)	(i) down 2; (ii) reflect in y-axis, asymptote $y = 1$	Reviewed	A y-axis reflection cannot move a horizontal asymptote
5 — 9	(a) completed square; (b) line; (c) region	(a) $12-\frac{4}{3}(x+2)^2 = -\frac{4}{3}(x-1)(x+5)$; (b) $y = -\frac{5}{4}x - \frac{25}{4}$; (c) three inequalities	**NOT reviewed live** [reference]	Keep the $4/3$; legacy guide dropped it
6 — 7	Simultaneous → quartic, all working	$(2x^2 + 5)(x^2 - 10) = 0$; $(\pm\sqrt{10}, \pm 4\sqrt{10})$	Reviewed (extra teaching)	$u = x^2$; reject negative u; exact surds
7 — 9	$f'(x) = 2x^{-1/2} + Ax^{-2} + 3$, $f''(4) = 0$; then $f(x)$ with $f(12) = 8\sqrt{3}$	$A = -4$; $f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}$	Reviewed	Method must justify; keep constants exact
8 — 10	Sector/arc: θ, arc, perimeter, area	$\theta = 5/3$, arc 35.9, perimeter 603, area 4730 cm² $= 0.473$ m²	**NOT reviewed live** [reference]	Radians; add both radii; cm²→m²
9 — 8	(a) trig in p; (b) sketch sin $2x$; (c) solve sin $2x = p$	(a) $2p, -p, 3 - p$; (b) period $180°$; (c) $\alpha/2,\ 90° - \alpha/2$	Reviewed & re-taught	Period $180°$; two roots by symmetry
10 — 12	Normal to curve, find k	(a) $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$; (b) $12x^2 + 4x - 33 = 0$; (c) $x = 3/2$; (d) $k = 5/4$	**NOT reviewed live** [reference]	Normal = negative reciprocal; algebra for k

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10. Conflicts resolved & score reconciliation

10.1 Authoring corrections (legacy guide → evidence-led rebuild)

- **Q5(a):** legacy guide printed $12 - (x + 2)^2$, which equals $-x^2 - 4x + 8$ and is **not** equal to the rest of its own chain (at $x = 1$: 3 vs 0). The correct first form keeps the $4/3$ coefficient: $12 - \frac{4}{3}(x + 2)^2$. ✓ fixed in the passed guide.
- **Q8/Q10 "NOT reviewed live":** the printed questions are present on the shared worksheet (frame-confirmed) but no working/solving is shown and audio has 0 hits on sector/arc/normal/"k" → **printed, not solved live**. Scored state unknown, not assumed blank.

10.2 Score — read carefully

The legacy **"29/75" is NOT a reconciled fact**. The per-question breakdown sums to **≈32/75** with a declared **≥3-mark unresolved delta** (Q8/Q10 scored state unknown; Q3/Q5/Q6 "essentially full" with no evidence of further loss). **Do not publish "29/75" as reconciled.**

Q	Max	Best-available earned	Basis
Q1	4	1	lost ≈3 (integration)
Q2	5	4	lost ≈1 (4-d.p. accuracy)
Q3	5	5	"essentially full"
Q4	6	3	lost ≈3 (transformation)
Q5	9	9	"essentially full" (reference)
Q6	7	7	"essentially full" (extra teaching)
Q7	9	3	lost ≈6 (differentiation)
Q8	10	0	unknown — NOT reviewed live
Q9	8	0	lost ≈8 (trig)
Q10	12	0	unknown — NOT reviewed live
Total	**75**	**≈32**	with declared ≥3-mark unresolved delta

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11. Exam-technique & notation jargon (whole paper)

Term	Meaning	This paper	Trap
Show all stages	Calculator-only scores nothing	Q6 bars calculator-only	Answer only
Exact value	Surd, no decimal	Q3, Q6	Rounding
Accuracy mark	Precision stated (e.g. 4 d.p.)	Q2(a)	Round early
4 s.f. / d.p.	Count from first non-zero / after point	cross-paper	Confuse
Asymptote	Horizontal line a curve approaches	Q4	Move it on reflection
Radians vs degrees	Calculator mode per question	Q8	Mode wrong
$cm^2 \to m^2$	Divide by 10000	Q8(c)	State cm^2 as m^2

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12. Supplementary item (NOT a paper question)

ESAT indices warm-up `[AUDIO]`: simplify $\frac{9^{2n+1} \cdot 3^{4-3n}}{27^{n-1}}$. Write to base 3: $9 = 3^2, 27 = 3^3 \Rightarrow 3^{(4n+2)+(4-3n)-(3n-3)} = 3^{9-2n}$. **This is a supplementary ESAT warm-up, NOT a WMA11 Summer-2022 paper question.** Never present it as part of the exam.

Grounded in `guide_v2/` (passed independent AV audit v1) and its official QP/MS sources. No question, mark, value or tutor comment has been invented. Q5/Q8/Q10 are labelled NOT reviewed live; the score is ≈32/75 with a declared ≥3-mark unresolved delta — "29/75" is not asserted as reconciled. Not a release — internal recall aid.